BODY—POWER SLIDING DOOR OPERATION AND DIAGNOSTIC SERVICE TIPS	Article No. 02-18-4

FORD: 1999-2002 WINDSTAR

Articles **00-11-4, 99-25-1 & 99-3-6** are being republished in their entirety to update the Diagnostic Information.

ISSUE

This article is intended as a comprehensive diagnostic aid to be used when servicing Power Sliding Doors.

ACTION

This TSB has been designed as a diagnostic supplement for the Workshop Manual. The Workshop Manual should be considered the primary source of information for diagnostic procedures. Refer to the following Service Information for details.

SERVICE INFORMATION

NOTE

IF THERE IS ANY INDICATION OR KNOWLEDGE OF PRIOR COLLISION DAMAGE, INSURE PROPER REPAIR PROCEDURES WERE FOLLOWED TO ELIMINATE THE POSSIBILITY OF MISALIGNED BODY PANELS CAUSING CONCERNS WITH THE POWER SLIDING DOOR SYSTEM.

Refer to Figures 1 through 26 for complete Diagnostic Procedure Supplement.

OTHER APPLICABLE ARTICLES: NONE
SUPERSEDES: 00-11-4, 99-25-1, 99-3-6
WARRANTY STATUS: INFORMATION ONLY
OASIS CODES: 102000, 110000, 111000, 112000, 701000, 702000, 703000

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HOW TO USE THIS DOCUMENT

This TSB is intended as a comprehensive diagnostic aid to be used when servicing power sliding doors.

This TSB has been designed as a diagnostic supplement for the Workshop Manual. The workshop manual should be considered the primary source of information for diagnostic procedures. The symptom chart "Figure 2" is a Trouble-Shooting Matrix. The matrix is made up of (horizontal) rows, and (vertical) columns. The description of their contents and how to use this document is as follows:

Rows

There are 27 rows contained in this matrix. Rows 1-8 identify 8 possible, generic descriptions of concerns. Rows 9-27 identifies 18 possible diagnostic trouble codes (DTC). All rows contain a series of numbers leading across the matrix. Each number leads up to a possible root cause listed in a column at the top of the matrix. The numbers contained WITHIN the matrix are organized by row, relative to the description of the concern, or DTC they represent, and are listed in order of ease to repair and the likelihood to occur. When the same number is repeated more than once in a row, it means that root causes identified by this number should be treated equal, and the repair procedures should be performed together.

Columns

There are 30 columns contained in this matrix. Each column has a number at the very top. These numbers identify the location of the repair procedure for the root cause listed below it. Repair procedures for root causes 1-15 are contained in the text of this TSB. The root causes listed at the top with a * above it are included here for identification purposes only, and already have adequate repair procedures listed in the Workshop Manual. You will need to refer to section 501-03-A of the Workshop Manual.

Trouble Shooting Matrix

To use this matrix, first identify the customer's complaint (left side) and choose the symptom that most closely describes the complaint, or DTC retrieved. The symptoms and DTC's are listed horizontally on the left side of the chart.
The possible causes are listed vertically, and identified by numbers in the order of which they are easiest to check or most likely to occur. The numbers that identify the root causes at the very top of the page, correspond with the procedures contained in the following document.
If there are trouble codes present, follow the pin-point test in the Workshop Manual. If this does not resolve the problem or in the event that there are no trouble codes stored, refer to Trouble Shooting Matrix.

For example: The customer complains that the door pops open a few centimeters at the rear after closing. This corresponds to "Latched in secondary position – parked". Follow the #2 row across to find the # 1 in that row. This leads up to the easiest and most likely root cause, "Battery at low state of charge", which is procedure # 6. Find procedure # 6 in the TSB, perform the indicated repair and retest the system. If the problem persists, return to the symptom chart and go to the # 2 listed root cause for this concern in the causal order. The #2 on this row leads up to "Track fasteners not fully seated", which is procedure # 3. Continue on in this manner until the problem is corrected. It may be necessary to investigate each root cause or DTC identified for each concern, in order to prevent repeat repair attempts.

Pages 4-6 give an overview of system components and system operation. On page 6 is a list of Pre-diagnostic tips. It may be helpful to read through these before attempting diagnostics.

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Figure 1 - Article 02-18-4

REPAIR PROCEDURE NUMBER - TO BE USED WITH TSB TEXT

Windstar Power Sliding Door Cause and Effect Matrix

Description of Concern	Diagnostic Trouble Codes (DTC)	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	*	*	*	*	*	*
		Cable Damage - Stripped Frayed	Pad & Plunger - Dirty	Track Fasteners Not Fully Seated	Detent Switch Signal - Always On	Detent Switch Signal - Too Short	Detent Switch Short to Ground	Detent Switch Open Circuit	Short to Battery on Detent Switch	Pad & Plunger Internal Continuity	Latch Actuator Failed or Stuck Open	Latch Actuator - Stuck Closed	Battery at Low State of Charge	Cable & Guide Interference - 4mm Hinge	Pad and Plunger - Ground Return	O/S Handle - Misinstalled	I/S Handle - Misinstalled	Hair Trigger Latch/Detent Switch	Manual Hinge Bumper on Power Hinge	Latch Actuator Circuit Open	Hinge and Sheet Metal Interference (LH)	PSD Controller	Door Adjustment - Door Fit	Optical Sensor	Fuel Filler Door Circuit	Parked Signal Open Circuit	RKE Shorted to Ground	B-Pillar Switch Shorted to Ground	Console Switch Shorted to Ground	VSS Signal Open Circuit	Console On/Off Switch Open Circuit
1	Unintended Door Reversals	3	1	2	6	6	6	6	6	6	7			7	8	4	4	5	5								10				
2	Latched in Secondary Position - Parked	4	3	2			3	7	6	6		7	1	7	8		1	2		5		6	9				6	9	5	4		
3	Door Opens by Itself - No Command															1	1							3							
4	Door Stops in Mid-Travel	2										7	1																		
5	Door Jerks Back Open from Closing Direction																					2		1							
6	Door Power Opens Approximately 4 in. and Stops																			8				1					5	10	
7	Door Will Not Power Open		2		7		7	7	7	7		9	1		6										3	4					
8	Noises from Power Door																					1									
9	B1342 ECU Defective																														
10	B2238 Brocken Cable Detected	1																													
11	B2592 Door Not Pulled into Primary During Power Close and High Duty Cycles	3		2									1	6					4			5	7								
12	B2374 Latch Switch Input Short to Ground				1		1																			1					
13	B2594 No Movement Detected After an Unlatch During Power Open																							1			1				
14	B2589 No Latch Switch Sensed on Closing and Door Reversed		1		3	3	3	3	3	3	5				2	2	4														
15	B2364 Fuel Filler Door Open Circuit																									1					
16	B2483 Park Signal Open Circuit																										1				
17	B2362 Remote Keyless Entry PSD Signal Shorted to Ground																											1			
18	B2365 B-Pillar Open/Close Switch Shorted to Ground																												1		
19	B2366 Driver Open/Close Switch Shorted to Ground																													1	
20	B2605 VSS Signal Open Circuit																														1
21	B2604 Console On/Off Switch Open Circuit																														1
22	B2363 Optical Sensor Failure																							1							
23	B2591 No Latch Switch Detected on Unlatch		1		4	4	4	4	4	4		5					2	2	3		6									6
24	B2593 Door Reversed While Closing Due to Obstacle Sensed	3	1	2	6	6	6	6	6	6	7						4	4	5												
25	B2603 Door Not Pulled Into Primary Under Power Close During Self Test	3	7	2						8				1	6					4		5	9								
26	Door Took Too Long to Cinch Into Primary From Secondary	3	7	2										1	6					4		5	9								
27	B2271 Door Did Not Reach Full Open	2		1																			4		3						

* Adequate procedures available in Workshop Manual Section 501-03.

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Figure 2 - Article 02-18-4

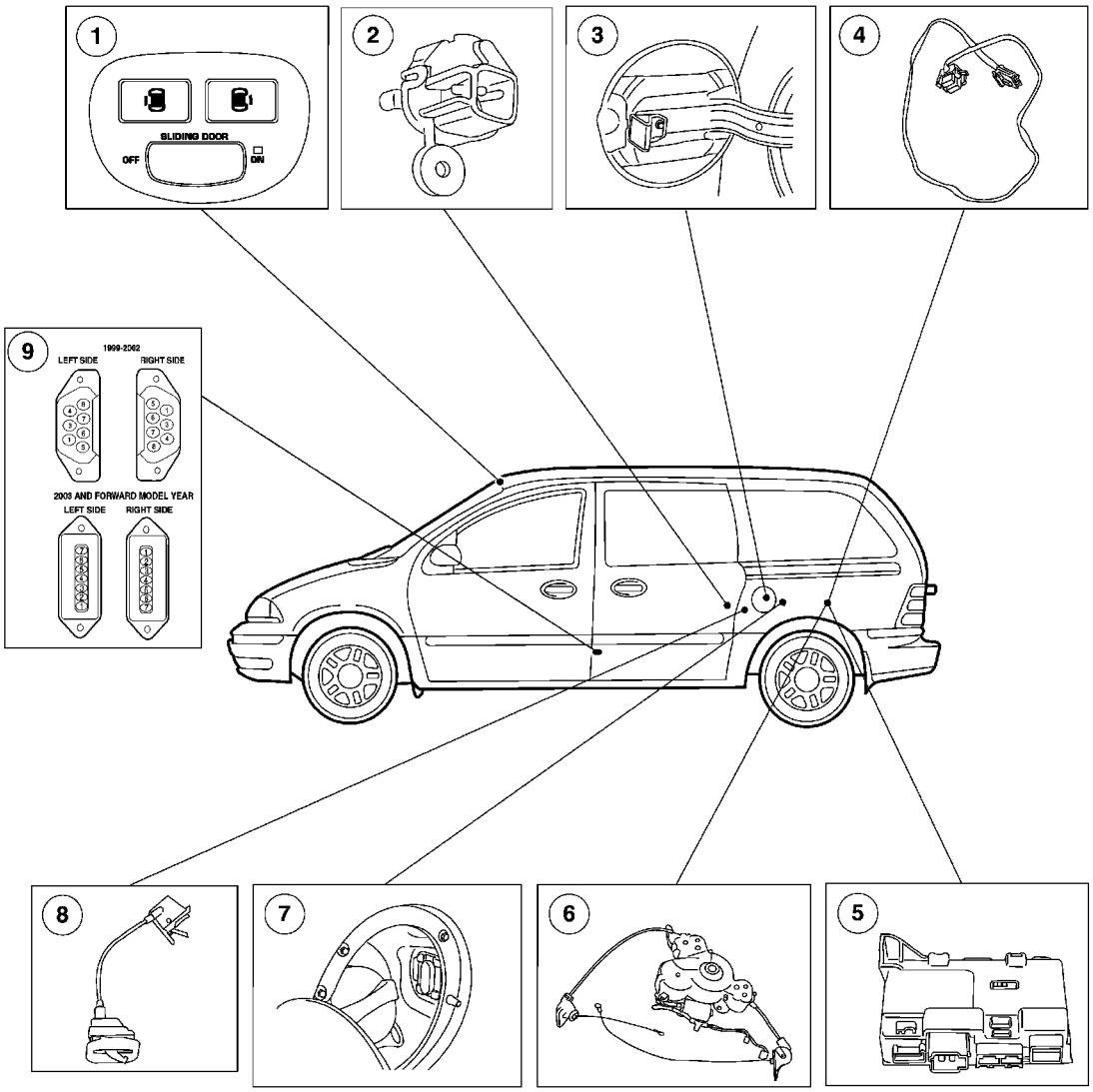
PRE-DIAGNOSIS TIPS

1. Make sure the battery is fully charged. The power sliding door modules are electronic devices. In order to perform logic functions correctly and apply appropriate power to the motors, the modules require maximum available voltage.
2. Ensure that the customer is not operating the doors for extended periods of time while the vehicle is not running. This will deplete battery voltage available and may affect the operation of the doors.
3. Ensure that the customer is not operating the door while the vehicle is parked on a steep incline (30 + degrees). Though the door is electrically controlled, the force of gravity and natural flex of the vehicle body may affect the proper operation of the door.
4. The force required to reverse the movement of the door are not adjustable. Factors that may affect this are parking on an incline, low battery voltage, and a dirty or otherwise affected track.
5. When the vehicle has sat for over 30 minutes and the customer attempts to open the sliding door with the handle before otherwise "waking up" the vehicle, the door will not immediately power open. The REM is sleeping and therefore not able to provide the "park" and "speed" signals to the PSD module. By the time the REM is awake and the door ajar light is on, the detent signal has already been processed and the PSD module relies on the motion sensor to provide an "opening" input. When the door has been moved 1/3 to 1/2 of the way open, the door will power assist the rest of the way.
6. The doors are not identical. The right door is larger than the left, and some customers may complain that they do not open at the same rate, close at the same rate, or they may have to pull one door further than the other before power assist takes over. Due to the different sizes and weights of the doors, this is normal and cannot be changed.

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Figure 3 - Article 02-18-4

OVERVIEW OF THE POWER SLIDING DOOR SYSTEM



ITEM	DESCRIPTION
1	Overhead console ON/OFF switch
2	Power sliding door (PSD) latch switch
3	Fuel filler door magnet
4	Right position sensor Left position sensor
5	Control module
6	Right drive assembly actuator Left drive assembly actuator
7	Fuel filler door switch
8	Fuel filler interlock catch
9	Pad and plunger and connectors

(A BASIC DESCRIPTION OF COMMON COMPONENTS FUNCTIONALITY OF THE SYSTEM ON FOLLOWING TWO PAGES)

TB-6771-A

Figure 4 - Article 02-18-4

Sliding Door Module
The power sliding door is a module-controlled system. The module controls the power sliding door activation depending on the position of the overhead console ON/OFF switch, the input received, the position of the door, and the direction of door travel.

Drive Assembly Actuator
The power sliding door drive assembly actuator consists of a motor, clutch and cable drive assembly. The actuator uses a cable-drive system connected to a power sliding door center roller hinge to open and close the door. The actuator and cables are installed and serviced as an assembly.

Pad and Plunger Assembly
The pad and plunger creates electrical connections across the door opening to provide power to the lock circuit, the power sliding door latch release actuator, and the detent switch. The pad and plungers are common to both sides of the vehicle but are rotated 180 degrees from side to side.

Latch Release Actuator
The latch release actuator is mounted on the remote control assembly inside the power sliding door, and is controlled by the power sliding door module. The actuator releases the door latch when a power open function is selected.

Position Sensor
The position sensor is located on the drive assembly actuator. The position sensor is used by the control module to record the position and velocity of the power sliding door.

Power Sliding Door (PSD) Latch Switch
The PSD latch switch is located on the rear sliding door latch. The switch identifies the latch switch position for open and close operations by supplying a ground signal to the power sliding door module.

Overhead Console On/Off Switch
The power sliding door overhead console ON/OFF switch is located in the overhead console. With the switch in the OFF position, the power sliding door will not operate from the B-pillar switch and power assist is disabled. Operation from the overhead console switch and from the key-fob is not affected by the ON/OFF switch position.

Power Assist
The power assist operation will operate only with the overhead console on/off switch in the ON position. Power sliding door opening and closing can be initiated by manually opening or closing the sliding door. From a fully latched position, power sliding door open is initiated by releasing the door latch with the interior or exterior door handle. If the sliding door is unlatched and partially open, the sliding door can be power opened by moving the door towards the open position. Power operation will take over after three to six inches of movement. Power closing is initiated by manually moving the sliding door towards the close position. After three to six inches of movement, the power close operation will take over and close the sliding door.

Power Sliding Door
The power sliding door is designed to open and close using:
- overhead console Open/Close switch
- B-pillar switch
- remote keyless entry transmitter (if equipped). Remote keyless entry activation requires the sliding door transmitter button to be pressed twice within three seconds.
- power assist

The power sliding door must be unlocked before the power sliding door open functions will operate (except for remote keyless entry activation). For the left power sliding door, the fuel filler door must be closed.

Power Sliding Door Operation

Opening operation may be stopped in four ways:
- Power sliding door reaches full open position.
- The open operation takes longer than ten seconds.
- The open operation was begun by a closing reversal and the door encounters an obstruction.
- If the power sliding door opening is obstructed before it reaches 30 cm (12 inches).

Closing operation may be stopped in two ways:
- The power sliding door reaches the full close position.
- The closing operation runs for longer than ten seconds.

When closing, the force required to reverse the action varies depending on the position of the door. During closing operation, the control module may interpret a gradual increase in resistance as a dirty track and will apply more power to the motor to overcome the resistance. This is a normal condition designed to compensate for a dirty or corrosive track or possible ice buildup during winter conditions. In order to reverse the closing operation, an abrupt or firm force against the door's direction of movement is required. Once the power sliding door is closed to the secondary latched position, the control module will apply full power to the motor in order to completely latch the power sliding door. In order to stall the closing operation within this distance, the opposing force or obstruction must be greater than that of the motor at full power.

Reversing Door Operation

During the power sliding door closing operation, travel can be reversed by any of the following actions:
- overhead console open/close switch
- B-pillar switch (if the overhead console switch is in the ON position)
- remote control keyless entry transmitter
- opposing force such as an obstruction

During the power sliding door open operation, travel can be reversed after the door opens approximately twelve inches by activating any of the following:
- overhead console open/close switch
- B-pillar switch (if the overhead console switch is in the ON position)
- remote keyless entry transmitter

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Figure 5 - Article 02-18-4

Repair Procedure 1

Cable damaged - stripped or frayed

a) Inspect front and rear cables for damage. The cable may be abraded or frayed, resulting in increased door opening/closing effort. The increased effort may trigger the obstacle detection and cause the door to reverse unintentionally. The cable may be frayed due to mis-assembly of the front and/or rear cable guide assembly.

Repair Action

b) Replace the drive unit. Pay close attention to the assembly of the cable guide to insure it extends through the opening so that the cable does not contact sheet metal while opening.

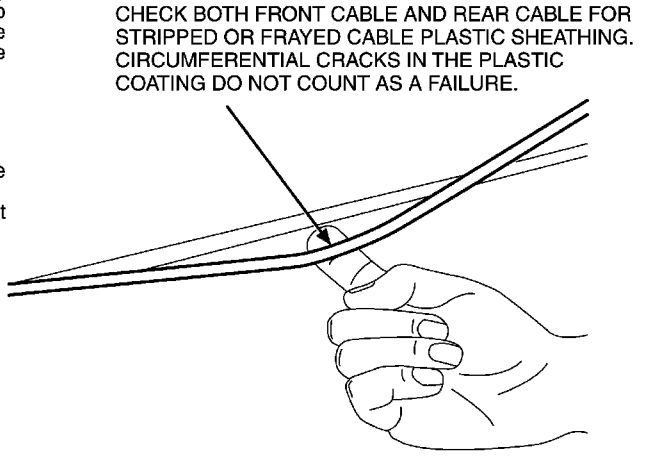


Figure 6 - Article 02-18-4

Repair Procedure 2

Pad and Plunger Dirty

a)Visually inspect for dirt and carbon build up, corrosion, wax, or any kind of coating tha can interfere with electrical contact. Also check for potential wear through of the outer plating on the contacts.

Repair Action

b) If it is dirty, use only a clean, dry cloth to rub the contacts clean. Do not use dielectric grease, sandpaper, emery cloth, or any abrasive cleaners.

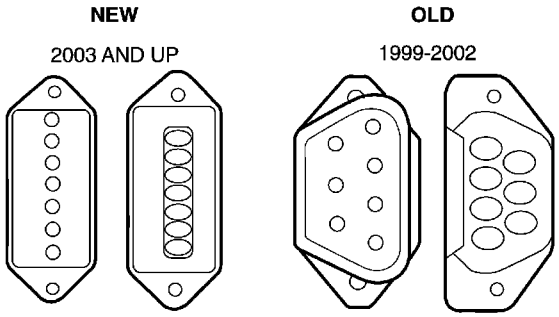


Figure 7 - Article 02-18-4

Repair Procedure 3

Center Track Fastener Not Fully Seated

a) Make sure the head of the fastener is seated fully ito the track.

Repair Action

b) Tighten and reinstall as needed. Ensure that the heads of each fastener does not restrict the movement of the hinge rollers.

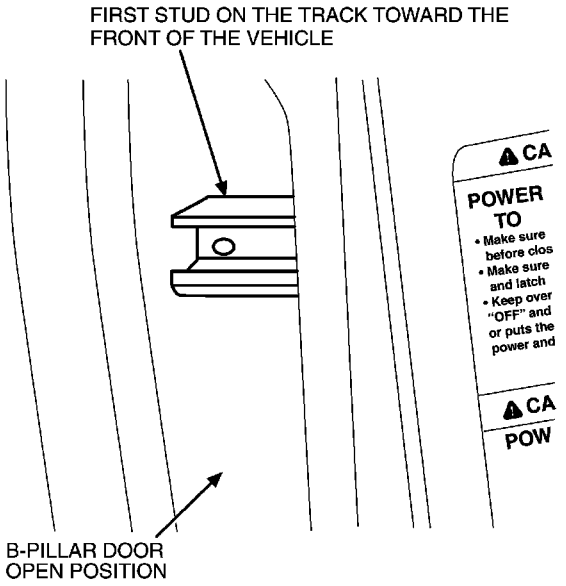
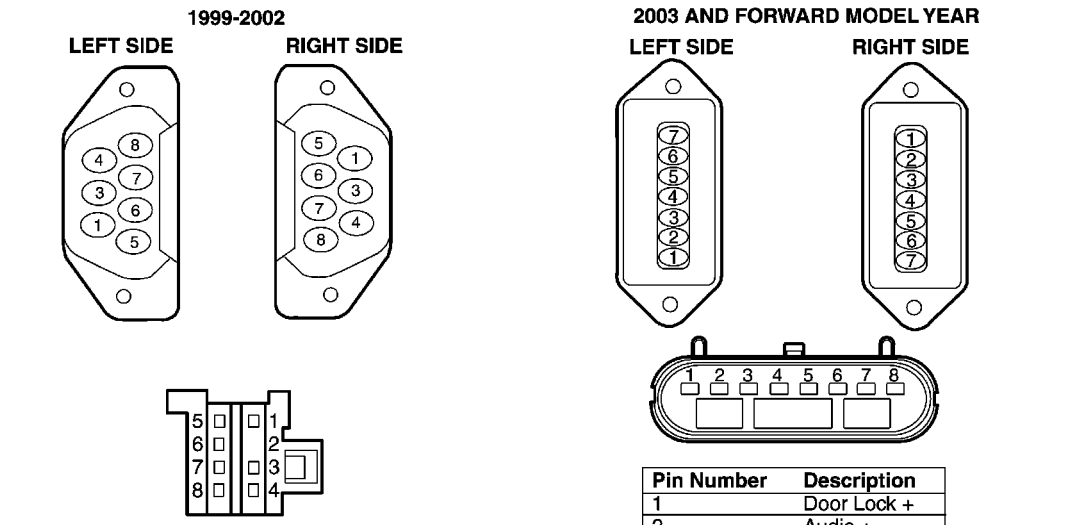


Figure 8 - Article 02-18-4

Repair Procedure 4

Detent Switch Circuit Diagnosis



Pin 1	Ground	Pin 2	No Connection
Pin 3	Latch Switch	Pin 4	Latch Actuator
Pin 5	Audio +	Pin 6	Door Lock +
Pin 7	Door Lock -	Pin 8	Audio -

Pin Number	Description
1	Door Lock +
2	Audio +
3	Door Lock -
4	Audio -
5	Ground
6	Detent
7	Latch Actuator
8	No Connection

Figure 9 - Article 02-18-4

Repair Procedure 4
(cont'd)

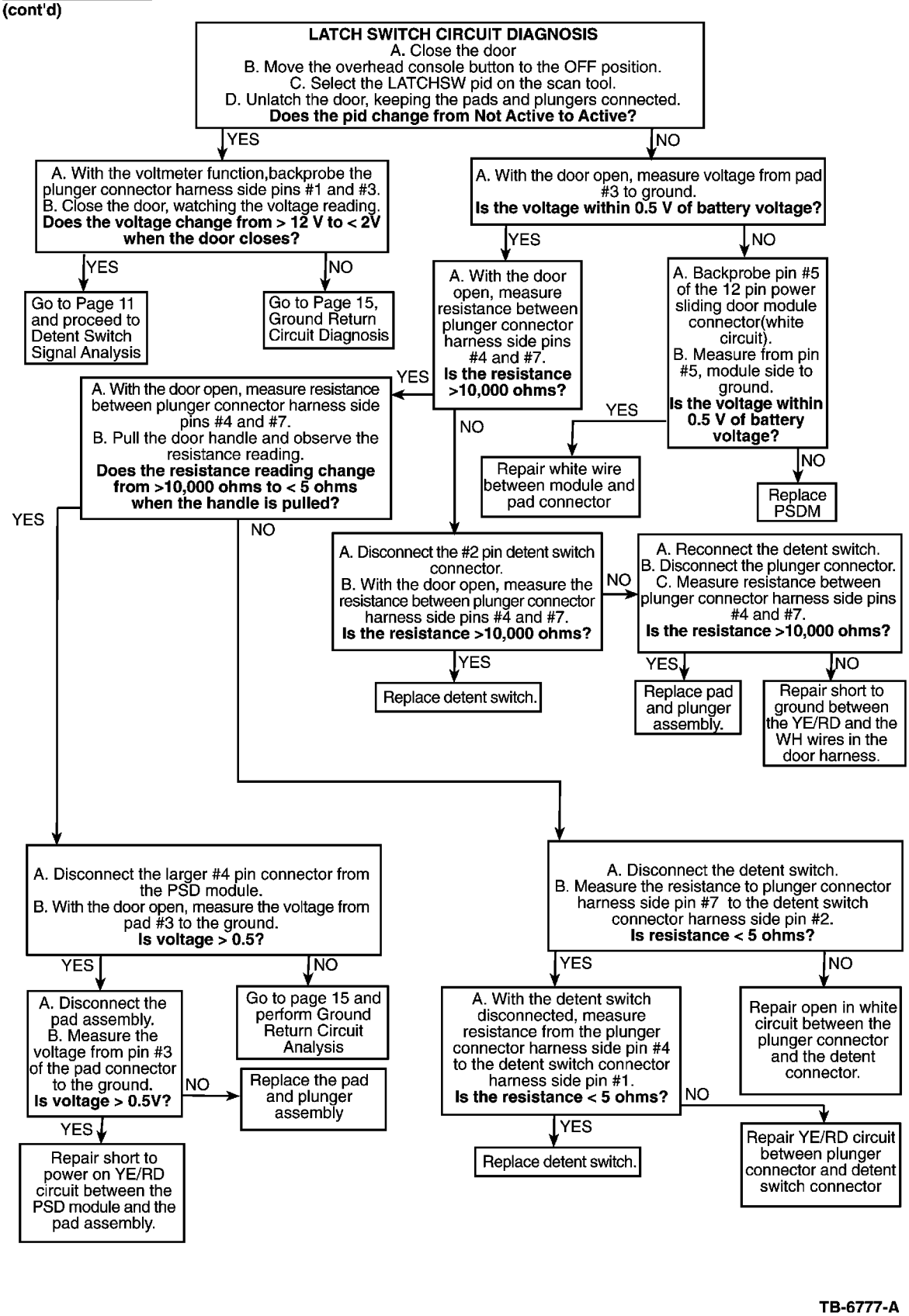


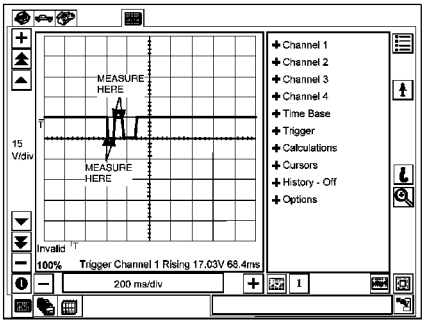
Figure 10 - Article 02-18-4

Repair Procedure 4
(cont'd)

WDS OSCILLOSCOPE-SETUP PROCEDURE

Note: When using WDS version B-21 and above, steps 1-10 can automatically be performed by entering "guided diagnostics" and selecting "PSD SWITCH".

1. Select The "Toolbox" icon at the top left of the screen.
2. Select "Oscilloscope" and press the "Tick" (check) mark.
3. Select "Channel 1" from the right side of the screen.
4. From the selections under "Channel 1", choose "Manual", "Sense", "Red and Black Probe Differential"
5. Select "Calculations" from the right side of the screen.
6. From the drop-down menu, select "Channel 1", then select "Frequency".
7. Set time base parameters by pressing the " – " sign at the bottom of the graph to 200 ms/div
8. By pressing the "-" on the left side of the screen, select 15V/div.
9. Press the oscilloscope icon with the "T" at the bottom center of the screen and move the trigger up _ a square by pushing the ^ button.
10. From the right of the screen, select "Cursors", "Channel 1", and an A and B button will appear to the right.
11. Hook the "B-203" cable to the PTU and the DLC.
12. Insert the red probe into the C1 port on the PTU and the black probe into the C2 port of the PTU.
13. Position the weather strip out of the way.
14. Position the left or right lower B-pillar trim panel out of the way, disconnect the contact pad connector inside the B-pillar, and back-probe the detent switch circuits 1251 (YE/RE) with the black probe and 1227 (WH) with the red probe. NOTE: USE ADAPTER TIPS FROM WDS KIT # 418-F264 OR EQUIVALENT. THE RED AND YELLOW MALE PIN ADAPTERS ARE RECOMMENDED.
15. Reconnect the contact pad connector and connect the red and black probes to the adapter tips.
16. Start the engine and activate the oscilloscope by pressing the icon of the man.
17. Operate the sliding door. As the door starts to latch, waveforms will appear on the screen.
18. When the waveforms appear, touch the icon of the running man to pause the screen. If two waveforms are not on the screen, re-activate the scope, operate the door, and pause the screen as the waveforms appear.
19. Push A, then push the fast forward button at the bottom of the screen to move the cursor to the desired position.
20. Push the B button and then use the fast forward button at the bottom of the screen to move the second cursor. At the center bottom of the screen, the time in milliseconds between the two cursors will be displayed.
    Read the secondary to primary latch time in milliseconds that is displayed at the bottom of the screen.



Check Timing of Detent Switch Signal

4.1 Waveform should appear as figure above. If not, go to step 4.2.

a. Check the voltage reading at the point that the detent switch is active (i.e. during the time the signal goes to ground.
b. Measure the time from the first falling edge to the first rising edge, then from the first rising edge to the second falling edge of the signal waveform
c. If either time is < 50 milliseconds replace detent switch, otherwise detent switch circuit is OK. Return to Symptom chart diagnostics.

4.2 Check for corrosion, or intermittent concerns

a. Re-check/re-clean pads and plungers.
b. Inspect harness side and component side of connectors for pads and plungers for corrosion or for pushed out terminals, or spread terminals (harness side).
c. Inspect harness side and component side of detent switch connector for corrosion or for pushed out terminals, or spread terminals (harness side).
d. Inspect door harness between detent switch and plunger assembly for abrasion, trapped/pinched wires, or cut wires.
e. Inspect harness side and component side of 4 pin PSDM connector for corrosion or for pushed out terminals, or spread terminals (harness side).
f. Body harness between pad assembly and PSDM for abrasion, trapped/pinched wires, or cut wires.
   Repair as necessary.

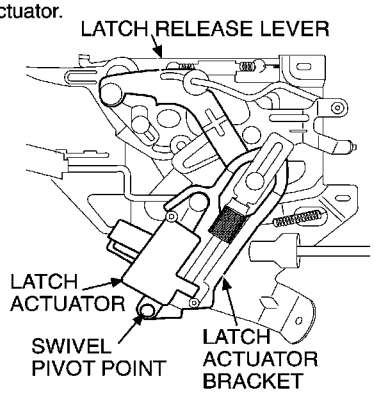
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Figure 11 - Article 02-18-4

Repair Procedure 5

Latch Actuator Failed or Stuck Open

a)Remove trim panel and examine latch actuator. To verify failure, turn the PSD overhead console switch to the off position. Actuate the door I/S or O/S door handle, and while holding or maintaining the handle in the actuated position, manually cycle the actuator by hand. If the actuator binds, replace the latch actuator.



NOTE: RELEASE THE I/S OR O/S HANDLE AND CHECK TO ENSURE THE ACTUATOR AND ITS BRACKET ARE FREE TO SWIVEL. IF SWIVEL BINDS, REPLACE ACTUATOR.

Repair Procedure 6

Battery at Low State of Charge

a) Check Battery Voltage. If less than 10.5 volts, charge battery.
b) Retest door.

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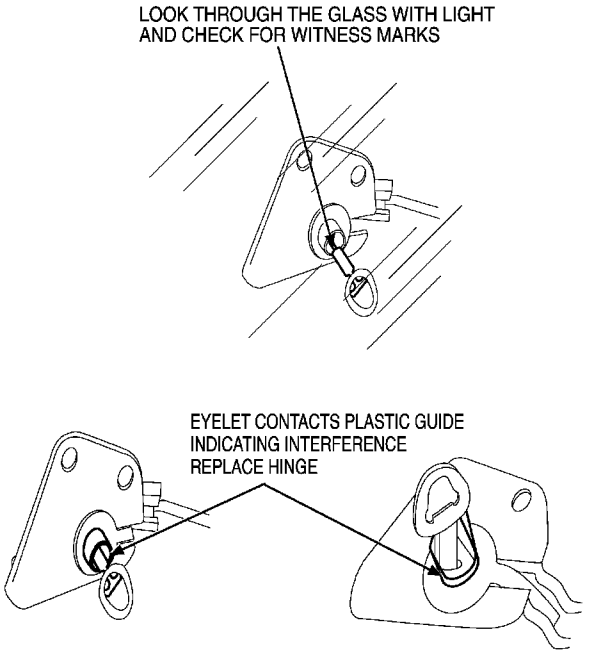
Figure 12 - Article 02-18-4

Repair Procedure 7

4mm Hinge Procedure

a) Check for interference between the cable-eyelet and the cable guide.

b) Unbolt the rear center hinge, lift the door upward and outboard to gain access to front guide assembly, and rest the door on a suitable support. Inspect the guide assembly for witness marks due to interference (scratches or gouges). If witness marks are noticed, replace the hinge. The current service hinge has been revised to allow additional movement of the cable and cable guide.



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Figure 13 - Article 02-18-4

Repair Procedure 8

Ground Return Circuit
Follow chart.

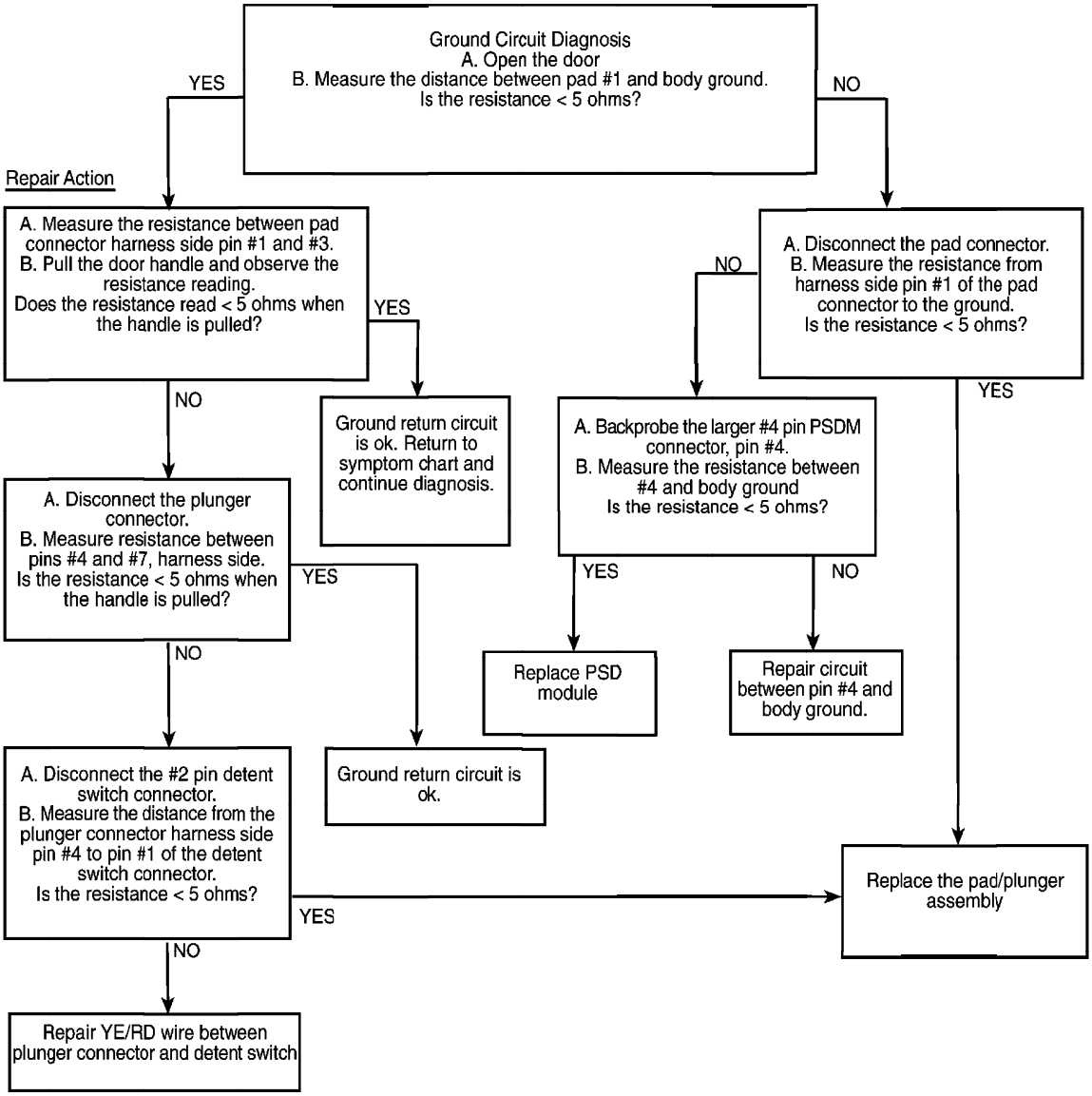


Figure 14 - Article 02-18-4

Repair Procedure 9

Inside or Outside Handle Mis-Installed

a. Check for preloading of the latch by the inside or outside handle.
b. Remove sliding door trim panel.
c. Inspect release lever position relative to the release lever stop (see figure 1 below).
d. The release lever should be resting on the release lever stop, with no gap between them. If a gap is present, preloading exists.
e. If preloading exists, disconnect both the inside handle rod AND the outside handle rod to relieve the preloading.
f. Reconnect the inside handle rod to the brain-plate, making sure the release lever is resting on the release lever stop.
g. Push the outside handle rod to the position shown in figure 2 making sure the release lever is resting on the release lever stop.
h. Reconnect the outside handle rod to the outside handle.

Figure 1

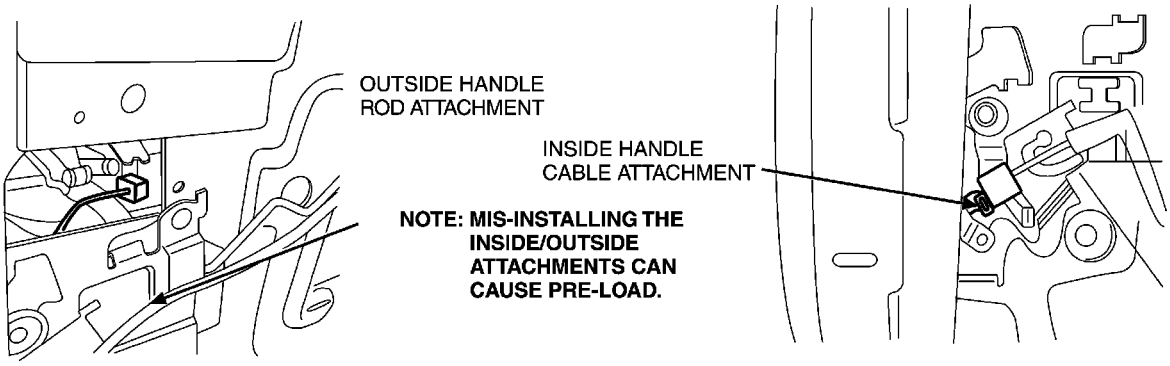
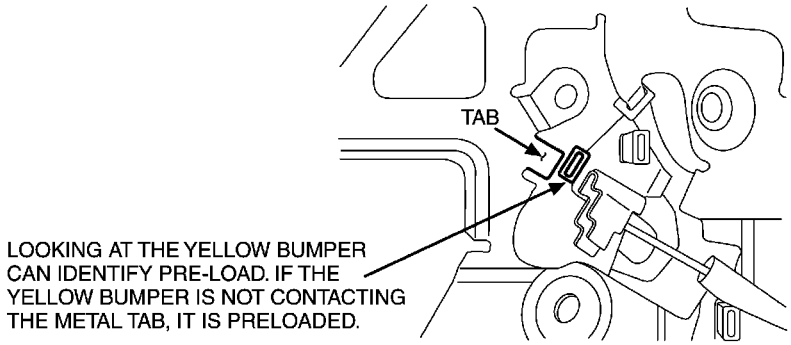


Figure 2



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Figure 15 - Article 02-18-4

Repair Procedure 10

Hair Trigger Detent Switch

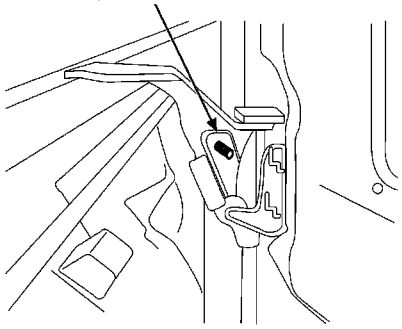
a. Close the door and turn system off using the on/off switch in the overhead console.
b. Read the LATCHSW PID using the WDS.
c. Slowly unlatch the door using the outside handle keeping the pads/plungers contacted until the LATCHSW PID reads ACTIVE.
d. Measure the door handle travel when the pid changes to ACTIVE. If the handle travel is < 4.5 mm, replace detent switch, otherwise detent switch function is OK.

Repair Procedure 11

Manual Hinge Bumper on Power Hinge

a) For the power sliding door sustem, the center hinge should not have a bumper attached. This bumper should only be present for the manual sliding door. If this bumper is present on a power door system, please remove.

HINGE BUMPER SHOULD NOT BE ATTACHED TO POWER SLIDING DOOR, IF PRESENT ON POWER HINGE, REMOVE.



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Figure 16 - Article 02-18-4

Repair Procedure 12

Latch Acutator Circuit

Follow Chart.

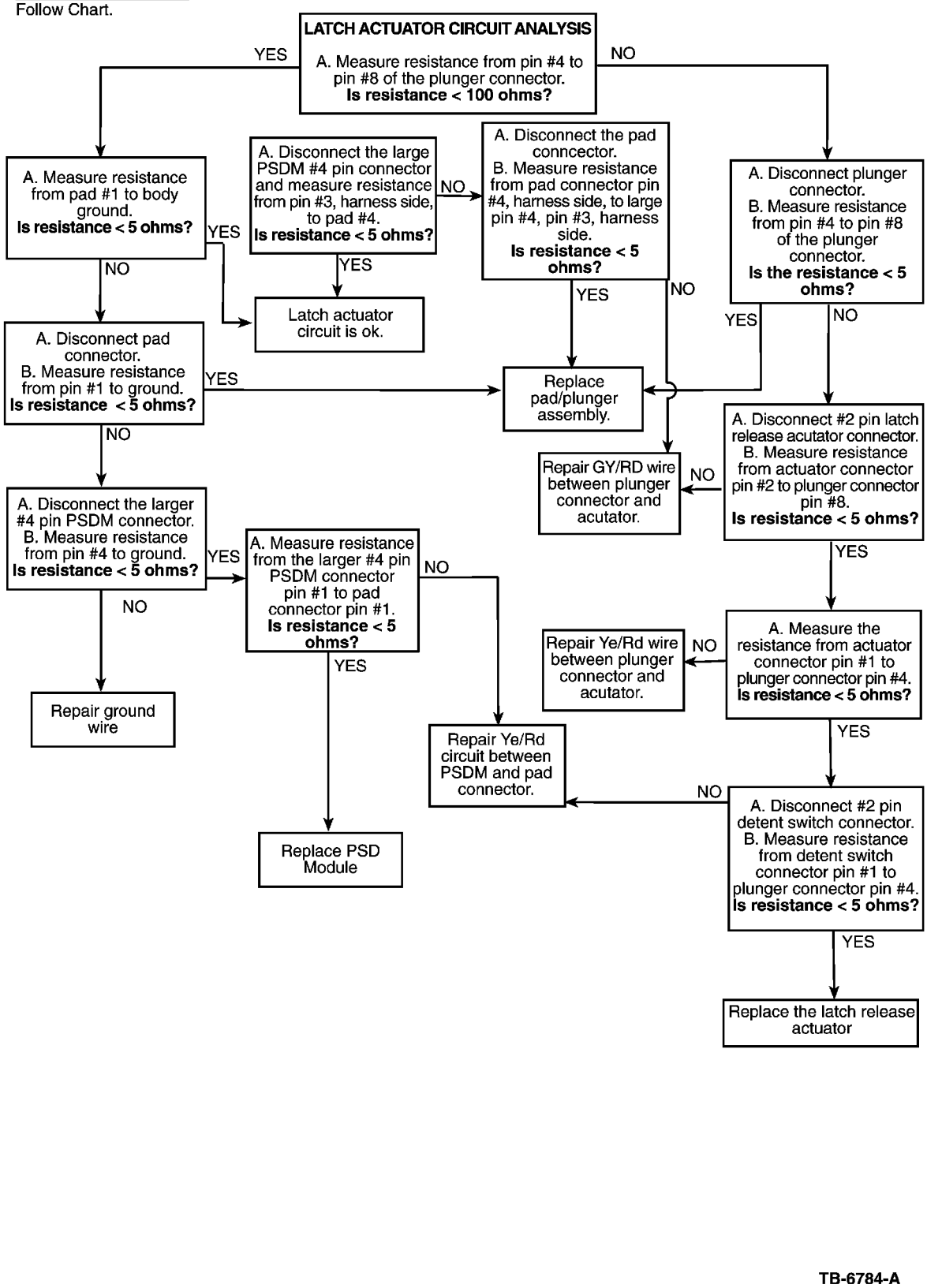
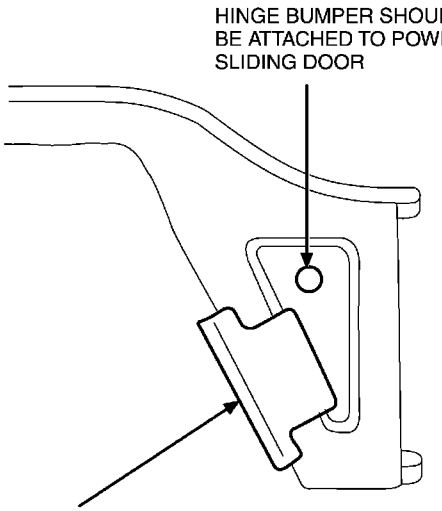


Figure 17 - Article 02-18-4

Repair Procedure 13

Hinge and Sheet Metal Interference

Attached to the center hinge body is a clip-on rubber bumper used to protect the paint through manufacturing only and serves no function after assembly. This bumper may contact the sheet metal with the door in the closed position causing higher closing forces. This bumper may be removed to improve door closing efforts



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Figure 18 - Article 02-18-4

Repair Procedure 14

Controller Assembly Relay -Copper Plating 1999 – 2000 MY Vehicles

On some vehicles, the power sliding door may intermittently not react to a closing command from any of the power sliding door switches or the remote key fob. Additionally, when attempting to close the door by pulling on the outside handle or pushing from the rear edge of the door, the door will resist the forward movement and will reverse back to the full open position. Relay contacts internal to the power sliding door controller for the 1999 – 2000 model year may be improperly plated and can build up an oxide, which results in an intermittent open circuit condition. The relays are not separately serviceable. Minor vibration of the power sliding door unit may temporarily shift the contact point and contribute to the intermittent condition. Frequent use of the power sliding door may clean the contact and prevent the oxide buildup. This condition is possible on all model years but is more visible with 1999 – 2000 model year vehicles. To service replace the affected power sliding door controller. Refer to the following Service Procedure for details.

SERVICE PROCEDURE

Replace the affected Power Sliding Door Controllers as appropriate. Vehicles that exhibit the condition may have the doors closed manually by turning off the power sliding door at the overhead console switch.

The power sliding door controllers have a five-digit date code on the white sticker, which has the part number on it. Parts which are date-coded "42329" and prior are more suspect to exhibit concerns resulting from improper internal plating, which causes internal oxide buildup. The second through fourth digits "232", refer to the day of the year (Julian date), the fifth digit is the year produced, "9" = 1999 and "0" = 2000. Comments for any claims submitted as a result of this symptom are as follows: RH/LH door will not close from full open position. Refer to the appropriate Workshop Manual Section for replacement procedure.

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Figure 19 - Article 02-18-4

Repair Procedure 15

Adjust the Sliding Door

The following procedure is used to repair windnoise, water leaks, and squeaks and rattles that are diagnosed as coming from the sliding door. The procedure consists of adjustments that are made to the sliding door to properly locate it in the sliding door opening.

a) Before any adjustments for fit are made, follow Steps 1, 2, 3 and 7 in Repair Procedure 15 - in that order.

b) Does "A" improve the alignment of the sliding door and stop the squeak and/or rattle? If YES, then the door is properly adjusted for minimizing squeak and rattle concerns. Retorque all bolts per specifications in Step 9. If NO, proceed to "C".

c)After performing "A", if the sliding door is still obviously misaligned for fit, follow Steps 1, 4, 5, 6 and 7 - in that order.

This guide was formulated by trained Windstar technicians, and is the plant's process for repairing the sliding door for squeak rattle concerns.

STEP 1 - LOOSEN FRONT WEDGES:

Use a chalk pen to outline the bolts on the wedges before they self adjust (Figure 1). This will give you an idea about how far out the sliding door wedges are when you first begin. If the wedges are not loosened (Figure 2), they will put the door in the same position the door was in before the adjustment was made.

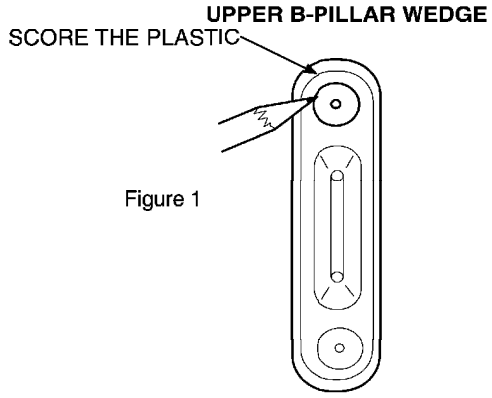


Figure 1

a) Loosen the wedge Torx bolts using a T-30 Torx bit.

b) Only loosen the wedge on the B-pillar; leave the receptacle on the sliding door secured.

The wedge should be loose enough to move, but not so loose that the wedge cannot hold its position when it is moved.

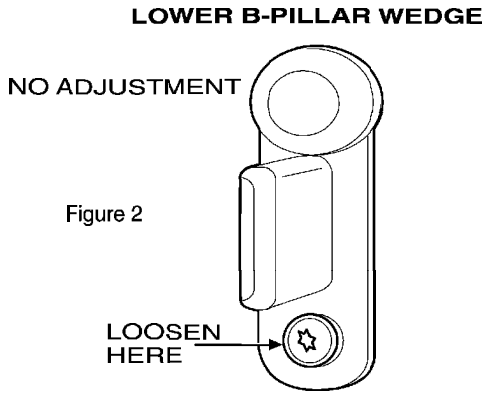


Figure 2

c) Loosen the wedge nut inside the B-pillar.

d) The wedge should be loose enough to move, but not so loose that the wedge cannot hold its position when it is moved.

e) Loosen the wedge Torx bolt using a T-27 Torx bit.

f) Again, only loosen the bolt slightly.

Once the wedge is snug/loose, the closure of the sliding door will realign the wedges to the proper position once adjustments are made for alignment and fit. Whenever adjustments to the sliding door are made, this procedure should be followed. The wedges are tightened in Step 7, after any adjustments are made to the sliding door.

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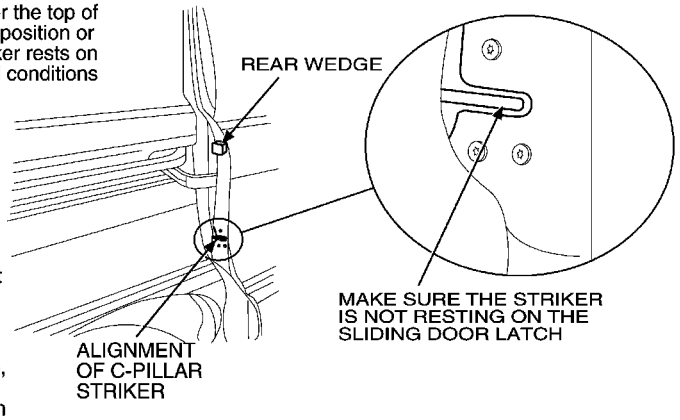
Figure 20 - Article 02-18-4

Repair Procedure 15 (cont'd)

STEP 2 - ADJUST C-PILLAR WIRE STRIKER:

The next adjustment to the sliding door is to check the alignment of the C-pillar striker

1. With the door all the way in the closed position, check the inboard/outboard up/down position of the striker to the latch. The striker should be sitting directly in the center of the latch for the up/down position.
2. The latch should NOT be resting on the bottom or the top of the striker when the door is almost in the closed position or in the closed position (see figure). When the striker rests on the top or bottom of the latch, vibration from road conditions will cause the door latch to rattle on the striker.
3. You can check the centering of the striker, by slowly opening and closing the door and watching the striker operate in the latch. The adjustment should be apparent after subjectively viewing the mating of the two components and any witness marks for wear on the striker.
4. Loosen the striker 1-3 turns so that it is snug but not loose. If the striker is loose, it will not stay in the position you adjust.
5. Up/Down Adjustment of the Striker: Again, the striker must be not be seated or touching the top, bottom or sides of the latch. Adjust the striker to the center up/down and inboard/outboard position by tapping the striker and checking its position after each tap.



NOTE: ONCE THESE STEPS ARE PERFORMED, ALL OTHER ADJUSTMENTS CAN BE REFERENCED TO THE C-PILLAR STRIKER. THIS STRIKER MUST BE ADJUSTED FIRST, THEN SUBSEQUENT ADJUSTMENTS CAN FOLLOW.

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Figure 21 - Article 02-18-4

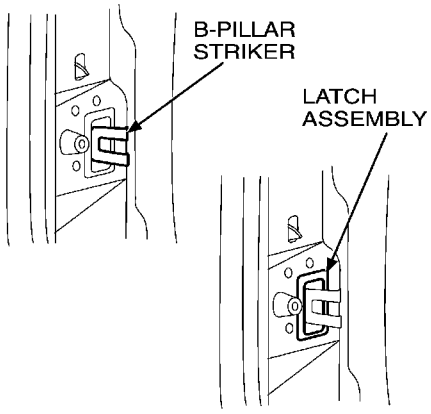
Repair Procedure 15 (cont'd)

STEP 3 - ADJUST B-PILLAR WIRE STRIKER:

With the door almost all the way in the closed position, check the inboard/outboard up/down position of the striker to the latch (see figure). The striker should be sitting directly in the center of the latch for the up/down position.

1. The striker should NOT be resting on the bottom or the top of the latch when the door is almost in the closed position and in the closed position. When the striker rests on the top or bottom of the latch, vibration from road conditions will cause the door latch to rattle on the striker.
2. You can check the centering of the striker, by slowly opening and closing the door and watching the striker operate in the latch. The correct adjustment should be apparent after subjectively viewing the mating nature of the two components and viewing any witness marks for wear on the striker.
3. Inboard/Outboard Sliding Door at the B-pillar:
   a. If the fit of the sliding door is inboard when the sliding door latch engages the striker, the striker must be tapped slightly outboard.
   b. Conversely, if the fit of the sliding door is outboard when the sliding door latch engages the striker, the striker must be tapped slightly inboard.
4. Up/Down Adjustment of the Striker: Again, the striker must be not be seated on the top or bottom of the latch. Adjust the striker to the center up/down position. This will prevent sliding door rattles.

NOTE: ONCE THESE STEPS ARE PERFORMED, ALL OTHER ADJUSTMENTS CAN BE REFERENCED BACK TO THE C-PILLAR STRIKER. THIS STRIKER MUST BE ADJUSTED FIRST, THEN SUBSEQUENT ADJUSTMENTS CAN FOLLOW.



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Figure 22 - Article 02-18-4

Repair Procedure 15 (cont'd)

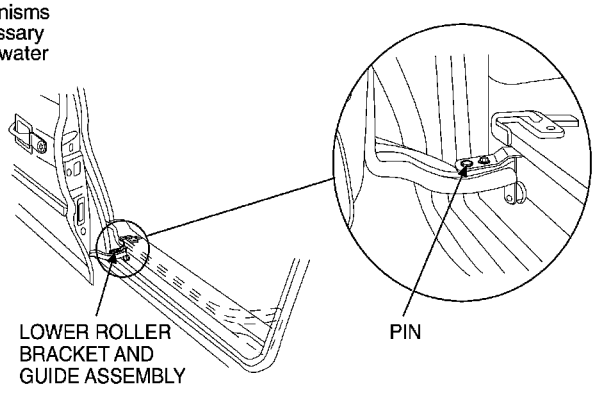
STEP 4 - LOWER ROLLER BRACKET AND GUIDE ASSEMBLY (LRBAGA):

NOTE: THIS ASSEMBLY CANNOT BE ADJUSTED FOR SOME VEHICLES IN 1998. IF THE PROCEDURE OF ADJUSTING THE STRIKERS FIXES THE RATTLE PROBLEM, ADJUSTING THE UPPER AND LOWER GUIDE ASSEMBLIES MAY NOT BE NECESSARY UNLESS THE FIT IS POOR.

After the wedges have been loosened, and the striker mechanisms have been corrected, the LRBAGA can be adjusted if necessary (see figure). This should only be attempted to adjust for fit, water leaks, and windnoise.

1. Loosen the bolt on the LRBAGA.
2. Move the bracket until the pin is centered in the window. Non-adjustable LRBAGAs do not have a window slot. If the assembly is non-adjustable, skip this Step.
3. Once the pin is centered in the window, the LRBAGA is properly adjusted.

If the pin is not centered in the window, the sliding door will not close flush with the body. The door will be too far in or too far out.



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Figure 23 - Article 02-18-4

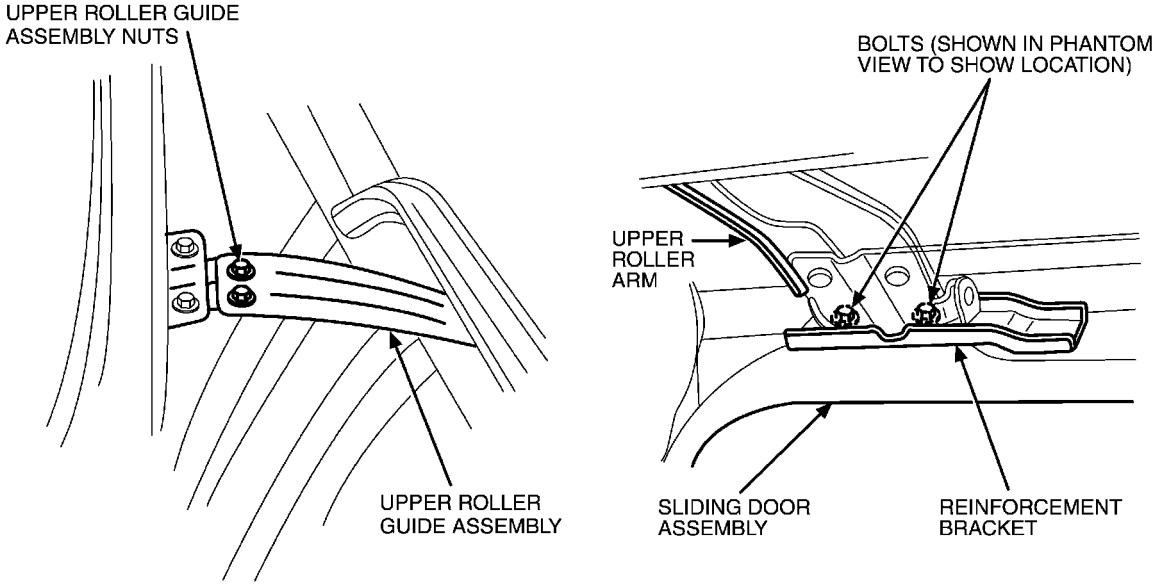
Repair Procedure 15 (cont'd)

STEP 5 - UPPER ROLLER GUIDE ASSEMBLY (URGA):

The URGA performs the following functions:

- ◆ Supports the top of the sliding door when it is open or closed
- ◆ It is one of the components that helps to close the door
- ◆ Provides proper door inboard/outboard positioning when the sliding door is closed.

After properly adjusting the LRBAGA, adjust the URGA (see figures 1 and 2 below). After the wedges have been loosened, and the striker mechanisms have been corrected, the URGA can be adjusted if necessary. This should only be attempted to adjust for fit, water leaks, and windnoise.



1. Mark the forward edge of the bracket so you can see how far you have moved the door.
2. Slightly loosen the nuts.
3. Move the door in the direction needed to make the door flush with the body. If the URGA is too far out, the following conditions may occur:
   - ◆ The door will not be flush with the body
   - ◆ The bottom of the door may be too far in
   - ◆ Windnoise at the upper front corner of the door
   - ◆ Water leak at the upper front corner of the door
   - ◆ Rattle at the upper front corner of the door.

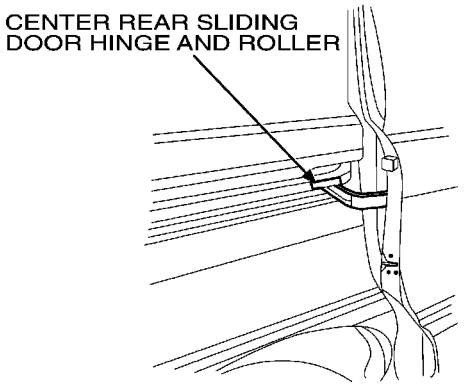
If the URGA was too far in, the door will not be flush with the body and the bottom of the door may be too far out.

STEP 6 - CENTER REAR SLIDING DOOR HINGE AND ROLLER:

The Center Rear Sliding Door Hinge and Roller Assembly (see figure below) is used to adjust the margin at the front and back of the sliding door. If the margin is incorrect at the front of the sliding door but within specifications at the rear of the sliding door, you will need to adjust the front passenger door. If the margin at the rear of the sliding door is incorrect, adjust the Center Rear Sliding Door Hinge and Roller.

CAUTION: DOING THIS WILL QUITE POSSIBLY RESULT IN THE MISALIGNMENT OF THE C- AND D-PILLAR STRIKERS. SACRIFICING FIT FOR RATTLE MAY BE THE ONLY CHOICE IF THE MARGINS ARE STILL OUT SLIGHTLY, BUT THE DOOR IS RATTLEPROOF.

1. Loosen the center rear sliding door hinge and roller bolts.
2. Adjust the door forward or backward to obtain the proper margin measurement.
3. Tighten the center rear sliding door hinge and roller bolts with the door closed.



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Figure 24 - Article 02-18-4

Repair Procedure 15 (cont'd)

STEP 7 - TIGHTEN WEDGES:

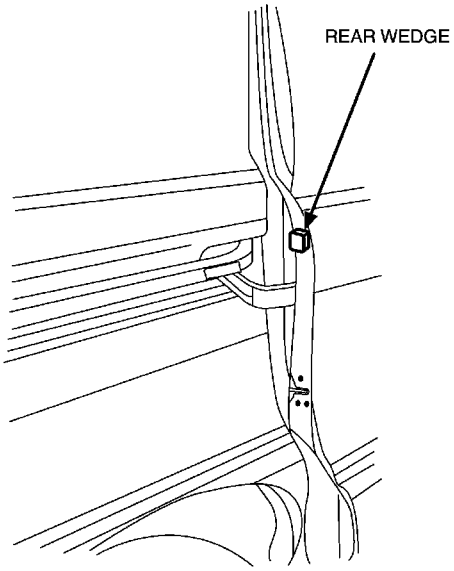
After completing all the necessary adjustments, the wedges can now be tightened.

1. The B-pillar male wedges should be snug and able to self-align when the door is closed.
2. Close the sliding door firmly to adjust the wedges to the right position.
3. Retorque the wedges to the proper torque carefully watching for movement while torquing. Any movement during retorque, start over.

The wedges should smoothly mate and NOT cause the door to pull inboard or outboard. The wedges are merely stabilizers that keep the door from movement in two axes, up/down and inboard/outboard.

4. Rear C-Pillar Wedge

The rear wedge (see figure) should engage smoothly as the door closes tightly. The plastic tape on the C-pillar should be aligned to contact the wedge smoothly as the door engages.



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Figure 25 - Article 02-18-4

Repair Procedure 15 (cont'd)

STEP 8 - Lubricate Check Bracket, Load Bearing, or Guide Bearing.

Grease the check bracket located at the back of the lower track. This bracket has a rubber pad brake that contacts the top/horizontal roller of the lower roller assembly located at the front/bottom of the sliding door.

PSD load bearing or guide may lack lubrication. Regrease the roller area with Silicone Lubricant or equivalent.

STEP 9 - TORQUE SPECIFICATIONS

TORQUE SPECIFICATIONS

COMPONENT	N•m	SAE
Bolt - B and C-Pillar Wire Striker	21-29 N•m	15-21 lb-ft
Bolt - Upper Roller Guide Bracket -to-Sliding Door	8-10 N•m	71-89 lb-ft
Nut - Upper Roller Guide Bracket-to-Adjusting Plate	8-10 N•m	71-89 lb-ft
Nut - Lower Roller Bracket and Guide-to-Adjusting Plate	8-10 N•m	71-89 lb-ft
Bolt - Lower Roller Bracket and Guide-to-Sliding Door	17-23 N•m	13-16 lb-ft
Bolt - Center Rear Sliding Door Hinge and Roller-to-Sliding Door	17-23 N•m	13-16 lb-ft
Nut - Sliding Door Track Check Plate	10-14 N•m	8-10 lb-ft
Nut - Body Side Center Track	5.5-6 N•m	48-53 lb-in
Bolt - B-Pillar Wedge	6-8 N•m	54-70 lb-in
Nut - B-Pillar Wedge (Inside B-Pillar)	6-7 N•m	54-61 lb-in

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Figure 26 - Article 02-18-4

NOTE: The information in Technical Service Bulletins is intended for use by trained, professional technicians with the knowledge, tools, and equipment to do the job properly and safely. It informs these technicians of conditions that may occur on some vehicles, or provides information that could assist in proper vehicle service. The procedures should not be performed by "do-it-yourselfers". Do not assume that a condition described affects your car or truck. Contact a Ford, Lincoln, or Mercury dealership to determine whether the Bulletin applies to your vehicle.